A host check for an STM32N657 CNN

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What was built

`stm32nn` is a CPU PyTorch extension for the host side of an STM32N657 export.

- The graded operation is `stm32nn.round_fp16`.
- The same module scores a network and prepares buffers before a flash.
- It is packaged as a Linux wheel: `CppExtension`, six pytest tests, `make`, `tools/smoke.sh`, and a two-stage Ubuntu 24.04 image.

`smoke.sh` prints `OK`. `docker run --rm fse-ops:0.1` prints `6 passed`.

The network

One MNIST CNN, trained 10 epochs on the PC.

- Two 3×3 convolutions, 32 base channels, two max-pools.
- Two linear layers, hidden width 128, dropout 0.25, weight decay 10^{-5}.
- 421 642 parameters. Best test accuracy 99.25%, final test accuracy 99.14%.

That checkpoint is the one loaded into the Helium firmware: FP16 weights, FP32 activations.

The graded operation

`round_fp16` takes a CPU float32 tensor and returns float32 of the same shape. Each finite element is rounded to the nearest IEEE binary16 value, ties to even, and stored back in float32. The specification line is

```
x.to(torch.float16).to(torch.float32).
```

An integer tensor raises `RuntimeError` from `TORCH_CHECK` in the C++ kernel. The six tests cover shape, dtype, an exact value, a value a copy of the input cannot survive, a leading batch of 4, and that error.

What the measurement used

- `round_fp16` cast every parameter of the CNN, then the network ran again in FP32.
- `compare` returns three numbers per pair of logit tensors: mean squared Euclidean gap, fraction of rows whose argmax differs, and mean cosine.
- `to_nhwc` / `to_nchw` match the channel-last buffers of a generated graph.
- `quantize_per_channel` is symmetric int8, scale $\max|w|/127$, codes in $[-127, 127]$, returned as float32.
- `check_export` is empty on this CNN with a zero 28×28 sample.

On the first convolution, that int8 rounding moves the weights by a mean square of 5.05×10^{-7}. That number is a weight change on the PC.

How the board run was made

The NUCLEO-N657X0-Q was on the ST-LINK. The Helium weight-only CNN was loaded into RAM and held running.

- MNIST test set, `numpy` seed 0, 200 images, indices sorted.
- Pixels divided by 255, sent as one NCHW float32 image (3136 bytes).
- The firmware cycle counter times the kernel. The serial transfer is outside that number.

The same 200 images were scored on the PC twice: stored FP32 weights, then every parameter passed through `round_fp16`. Cube.AI's FP32 graph is the saved serial run of this same subset.

Three scores, one subset

- Logit MSE: mean over images of $\|z - z_{\mathrm{FP32}}\|_2^2$. Ten pre-softmax numbers. This is not an average over the 10 classes.
- Cosine: mean over images of the cosine between those two vectors.
- Label accuracy: fraction of images whose largest logit is the written digit.

A large squared gap can sit next to a cosine near 1 and next to an unchanged winning class. The table keeps all three.

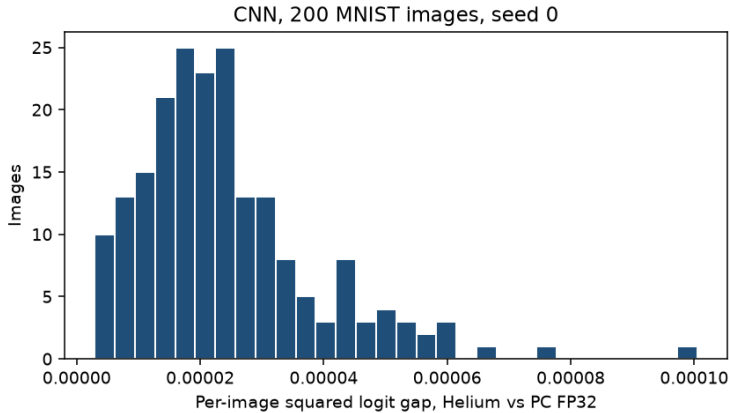
Results

Run	Logit MSE	Cosine	Class differs
PC, `round_fp16` on the weights	2.421720×10^{-5}	1.000	0/200
Helium on the board, 20 October	2.422037×10^{-5}	1.000	0/200
Cube.AI FP32, same 200 images	5.445340	0.998	0/200

Label accuracy is 0.995 on the untouched FP32 network and on all three rows: 199 of 200 digits.

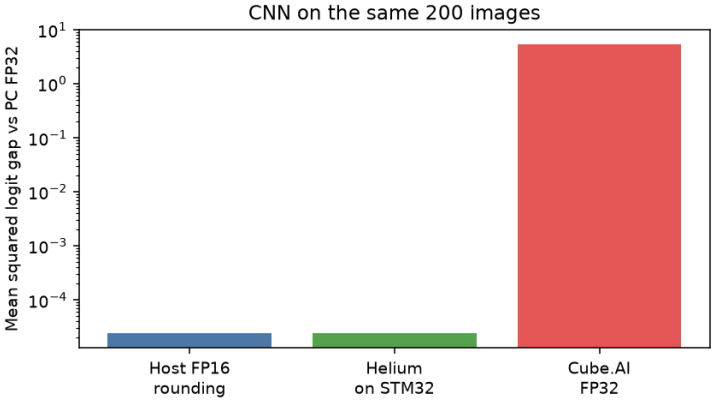
The eval script's own board file reports 2.422138×10^{-5}. `compare` and that file differ in the last digits.

Helium against the PC, image by image



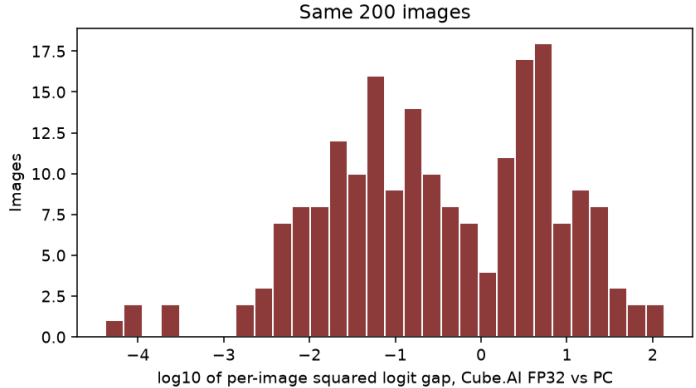
Every image stays small. Host rounding and the board differ by about 3×10^{-9} in the mean: 2.421720×10^{-5} against 2.422037×10^{-5}.

The three gaps on one axis



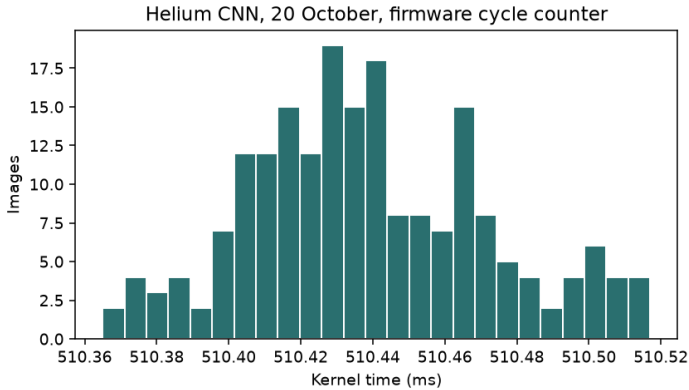
The left two bars are the FP16 rounding, on the PC and on the board. The right bar is the Cube.AI FP32 graph, about 2×10^5 times larger. The winning class still matches on all 200 images.

Cube.AI, per image



The axis is $\log_{10}$ of the squared gap, because the raw values span more than one decade. Cosine on this run is 0.998217. Argmax disagreement is 0/200.

Time on the board



Mean kernel time 510.4 ms (20 October). The 200 samples sit between 510.4 ms and 510.5 ms. About 1.96 inferences per second, serial time excluded.

Reading the table

- FP16 rounding of this CNN moves the logits by about 2.4×10^{-5} and changes no class on this subset.
- The Helium program, FP16 weights and FP32 activations, lands on that same number.
- Cube.AI's FP32 graph moves the logits by 5.445 and still picks the same class on every image. Cosine stays at 0.998.
- Label accuracy does not see any of that movement: it is 199/200 for FP32, for the host rounding, for Helium, and for Cube.AI.